

Lab: Thinking in Loops and Chains

Objective

Explore the cognitive dimensions of crochet by engaging in pattern reading, mental simulation, and the transition between automatic and controlled processing — all while observing your own thinking.

Materials

- Worsted weight yarn in a light color, approximately 30 yards
- Crochet hook, size H/8 (5.0mm)
- Two pattern cards (provided below — print or copy them before starting)
- Pen and Cognition Log (template below)
- Timer or phone
- A podcast, audiobook, or conversation partner

Pattern Cards

Pattern A (Simple): Row 1: Ch 21. Sc in 2nd ch from hook and in each ch across. (20 sc)

Rows 2-8: Ch 1, turn. Sc in each st across. (20 sc)

Pattern B (Complex): Row 1: Ch 22. Sc in 2nd ch from hook, *dc in next st, sc in next st*, repeat * to * across. (21 sts) Row 2: Ch 1, turn. *Sc in dc, dc in sc*, repeat * to * across. Row 3: Repeat Row 2. Rows 4-6: Repeat Row 2.

Cognition Log Template

Phase	Task	Could I think about other things?	Did I lose count?	Automatic or Controlled?	Notes

Steps

Part 1: Pattern Reading as Model Loading (10 minutes)

1. Read Pattern A without crocheting. Close your eyes and try to visualize the finished swatch. How clear is your mental image?
2. Now read Pattern B. Try to visualize this swatch. Is the image clearer or fuzzier than Pattern A? Why?
3. In your Cognition Log, record your observations. Which pattern was easier to "see" in your mind? This difference reflects the depth of your generative model for different stitch patterns.
4. Before crocheting, write a prediction: How will Pattern B's texture differ from Pattern A's? This is your generative model making a prediction.

Part 2: Dual-Task Test — Automatic vs. Controlled Processing (20 minutes)

1. Start crocheting **Pattern A** while listening to a podcast or audiobook (or having a conversation with a circle partner).
2. Time yourself for 5 minutes. At the end, note in your log: Were you able to follow the audio content while crocheting? Did you lose your stitch count? Did the crochet feel automatic?
3. Now switch to **Pattern B** with the same audio content.
4. Time yourself for 5 minutes. Note: Did you have to pause the audio or stop the conversation to focus on the pattern? Did you lose track of which stitch came next? Did the crochet demand more conscious attention?
5. Record the differences in your log. The shift you experienced — from easy dual-tasking with Pattern A to difficult dual-tasking with Pattern B — illustrates the transition from automatic to controlled processing.

Part 3: Mental Simulation and Counting (10 minutes)

1. Set aside the physical swatch. Without crocheting, mentally simulate working one row of Pattern B. Walk through it in your mind: "Sc in dc, dc in sc, sc in dc..." Count how many stitches you simulate before you lose track or your mind wanders.
2. Record this number. It is a rough measure of your internal simulation capacity for this pattern.
3. Now physically crochet one row of Pattern B while counting each stitch aloud.
Compare: was the physical version easier or harder than the mental simulation? Why?

Part 4: Prediction Testing (10 minutes)

1. Look at your Pattern A swatch and your Pattern B swatch side by side.
2. Compare them to the prediction you wrote in Step 4. Did Pattern B's texture match your expectation? Where was your generative model accurate and where was it off?
3. Feel both swatches with your eyes closed. Can you tell them apart by texture alone?
Record your observations.

Part 5: Reflection and Discussion (10 minutes)

1. Review your Cognition Log. Where did cognition feel effortless (automatic) and where did it require focused attention (controlled)? What triggered the transitions?

Discussion Questions

- When you read Pattern B for the first time, how much of the finished texture could you predict? What did you need to actually crochet before you could "see" it?
- During the dual-task test, what happened to your crochet when your attention was captured by the audio content? Did you make more errors? Did your tension change?
- When simulating stitches mentally, did you "feel" phantom hand movements? What does this suggest about the relationship between cognitive simulation and motor planning?
- How does this lab change your understanding of what "thinking" means during crochet?

Wrap-Up

You have explored crochet cognition through pattern reading (model loading), dual-tasking (automatic vs. controlled processing), mental simulation (internal counting), and prediction testing (model accuracy). These cognitive operations are at the heart of Active Inference — they are how your generative model engages with the world before, during, and between stitches.